Sample Paper 7 — Answer Key (Level-2 Difficulty Step-Up)

Total: 40 marks (30 × 1 mark + 5 × 2 marks), 60 minutes.

Section 1 — Logical Reasoning

1. **C) Potato** — Mango, Banana, and Apple are fruits; Potato is a vegetable.
2. **A) Cut** — A pen is used to write; scissors are used to cut.
3. **A) JDUGHQ** — The rule is "+3 letters" (wrapping Z→C): G→J, A→D, R→U, D→G, E→H, N→Q.
4. **C) 41** — Two patterns are interleaved: the odd-position terms go 3, 6, 9, 12 (+3 each); the even-position terms go 50, 47, 44, ? (−3 each) — so the next even term is 41.
5. **A) Nisha, Leela, Omar, Karan, Manoj** — Manoj is at position 5 (right end). Karan is immediately to his left, position 4. Leela is 2nd from the left, position 2. Omar sits between Leela (2) and Karan (4), so Omar is position 3. The only person left, Nisha, takes position 1.
6. **A) 10 m North** — East and West cancel (10 m each way); the two 5 m North legs add up: 5 + 5 = 10 m North.
7. **B) Saturday** — 28 − 5 = 23 days later; 23 = 3 weeks + 2 days, so the day of the week moves forward 2 days from Thursday: Saturday.
8. **D) 48** — 16 = 4×4, 25 = 5×5, 36 = 6×6 (each a number multiplied by itself — a "square number"); 48 does not fit this pattern.
9. **D) 360** — Each term is multiplied by one more than the last multiplier used (×3, ×4, ×5, ×6): 60 × 6 = 360.
10. **D) 16th** — The 3 new people join behind him, so 3 more people are now further from the front than Vikas: his position from the back increases by 3: 13 + 3 = 16th.

Section 2 — Mathematical Reasoning

11. **A) 82467** — 45678 + 36789 = 82467.
12. **A) 35433** — 70000 − 34567 = 35433.
13. **A) 16415** — 2345 × 7 = 16415.
14. **A) 588** — 8232 ÷ 14 = 588 exactly.
15. **C) 18** — $54 = 2\times3^3$, $72 = 2^3\times3^2$. HCF = $2\times3^2 = 18$.
16. **B) 45** — $9 = 3^2$, $15 = 3\times5$. LCM = $3^2\times5 = 45$.
17. **C) 11/12** — As decimals: 7/8 = 0.875, 5/6 ≈ 0.833, 11/12 ≈ 0.917, 4/5 = 0.8. The greatest is 11/12.
18. **A) 6 km 450 m** — 3650 m + 2800 m = 6450 m = 6 km 450 m.

19. **C) 640 sq m** — Area = 32 × 20 = 640 sq m.
20. **C) 12 cm** — A regular hexagon has 6 equal sides: 72 ÷ 6 = 12 cm.

Section 3 — Everyday Mathematics

21. **C) ₹17,100** — 68400 ÷ 4 = 17100.
22. **C) 1,250** — Rate = 2000 ÷ 8 = 250 toys/hour; in 5 hours = 250 × 5 = 1,250.
23. **C) 24 min** — 3000 ÷ 125 = 24 minutes.
24. **C) 4 h 30 min** — Total time 6:50 a.m. to 11:35 a.m. = 4 h 45 min; subtracting the 15-minute stop gives 4 h 30 min of actual travel.
25. **B) 1 L 500 mL** — 3600 mL + 2400 mL = 6000 mL; 6000 ÷ 4 = 1500 mL per bottle.
26. **C) ₹112** — 7 × ₹18 = ₹126; ₹126 − ₹14 discount = ₹112.
27. **C) 640 L** — 1/6 × 960 = 160 L used per day; over 2 days = 320 L used; 960 − 320 = 640 L remaining.
28. **D) 46 kg 500 g** — 12300 g + 15700 g + 9500 g + 8500 g = 46000 g = 46 kg.
29. **C) 35** — 945 ÷ 27 = 35 stacks.
30. **C) ₹320** — Sold for 40 × ₹48 = ₹1,920; profit = ₹1,920 − ₹1,600 = ₹320.

Section 4 — Achievers Section (2 marks each)

31. **B) 17** — Working backwards: (84 + 18) ÷ 6 = 102 ÷ 6 = 17.
32. **C) 20** — Let the first number be x. Then second = $2x$, third = $2x$+15. Sum: $x + 2x + (2x+15) = 115 \rightarrow 5x + 15 = 115 \rightarrow x = 20$.
33. **C) ₹2,000** — After spending 2/5 on books, 3/5 of the money remains. After spending 1/3 of that on stationery, 2/3 of 3/5 = 2/5 of the original remains. So 2/5 of the total = ₹800 → total = ₹2,000.
34. **C) 235** — Boys = (2 × 116) + 3 = 235. Check: 116 + 235 = 351. ✓
35. **C) 42** — Each term is the sum of the previous two: 16 + 26 = 42.

Scoring guide (out of 40, Level-2-difficulty paper): 28+ here is a very strong result — it's harder than Papers 1–5 by design, so the usual 32+ Level-1 benchmark doesn't directly apply. 20–27 is solid for a paper at this difficulty. Below 20 — review *why* each missed question was harder (an extra step, a bigger number, two ideas combined) rather than the raw score.